

# Direction-Invariant Auditing of Batch Correction in Biomedical ML, with Classical and Quantum Feature-Space Bounds

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## Abstract

We identify a specific failure mode of linear batch-correction *evaluation* in multi-cohort biomedical machine learning. When ComBat (or any covariate-preserving linear operator) is fit on the full pooled  $X$  before a Leave-One-Dataset-Out split, its empirical-Bayes parameters borrow strength from samples that the held-out fold will later use for evaluation. The corrected matrix inherits test-set information, and a downstream classifier appears to invert its decision direction – the apparent *label-flip*. We formalise the geometric condition under which a flip would arise on a properly fitted pipeline (**Theorem 1**, sharp phase transition at  $\rho^2 = \frac{1}{2}$ ), and operationalise it as the Direction-Invariant AUC Leakage metric (**DIAL**). We extend the analysis to quantum feature maps: **Proposition 2** shows that the flip probability under a uniform random quantum feature map in  $d = 2^q$ -dimensional Hilbert space is bounded by  $2^{-(d-1)}$ , an exponential suppression in the number of qubits  $q$ . We validate the diagnostic on controlled synthetic data (4,950 runs, real `inmoose.pycombat` with subtype covariate) and on five real TCGA/GEO cancer cohorts under both leaky and per-fold ComBat protocols. The leaky protocol reproduces a published “thyroid-specific” batch-entanglement pattern (THCA  $\Delta\text{DIAL} = 0.494$  on  $\text{LogReg-}\ell_2$ ); the per-fold protocol returns  $\text{DIAL} = 0$  on every cancer-classifier cell, including all five thyroid rows ( $\text{AUC}_{\text{post}} = 0.995$  on the same  $\text{LogReg-}\ell_2$  row). The reframed contribution is therefore: (i) DIAL functions as designed – it diagnosed the leakage rather than a biological flip; (ii) a per-fold ComBat protocol that prevents the inflation; and (iii) classical and quantum bounds on when a flip *would* actually occur on a correctly fitted pipeline. Qiskit 2.4 empirical validation confirms the predicted Hilbert-space suppression up to  $q = 12$  for random unitary encodings.

## 1 Introduction

Batch effects are a central obstacle in multi-cohort biomedical ML. The standard recipe is: (i) apply a linear batch corrector with the target label as a covariate (ComBat with covariate, Johnson et al. 2007); (ii) train a classifier; (iii) evaluate cross-cohort. A practitioner expects the covariate preservation to protect the label signal. We show two distinct failure modes.

**Mode A (theoretical).** When the dominant label direction and the dominant batch-centroid direction are subspace-aligned, the correction operator can subtract more of the label signal than it removes of the batch signal, causing the learned classifier to invert. Theorem 1 gives the sharp threshold.

**Mode B (the actual failure we observed).** Even on data that does not satisfy the alignment condition of Mode A, an apparent flip can arise from a much more mundane cause: the

empirical-Bayes batch correction is fit on the full pooled training + test matrix before the cross-validation split, so the correction parameters carry test-set information into training. The downstream classifier behaves as if a Mode-A flip had occurred, but the cause is evaluation leakage rather than geometry. In our motivating example (thyroid BRAF-vs-RAS), the leaky protocol reports  $\Delta\text{DIAL} = 0.494$  on the LogReg- $\ell_2$  classifier; the per-fold protocol on identical data returns  $\text{DIAL} = 0$  ( $\text{AUC}_{\text{post}} = 0.995$ ). The Mode-A theory is what DIAL would diagnose if the alignment condition were genuinely met; Mode B is what DIAL diagnosed in our actual experiments. Both are useful: Theorem 1 tells us when a flip is biologically expected, Proposition 2 bounds it under quantum feature maps, and the per-fold protocol of Section 6 distinguishes Mode A from Mode B in practice.

We introduce the *Direction-Invariant AUC Leakage* metric, **DIAL**, which flags this failure using cross-cohort AUC alone (§4). We prove a sharp phase transition at  $\rho^2 = 1/2$  (Theorem 1, §4.2) and provide an exponentially-tighter bound when the analysis is lifted to a quantum feature space (Proposition 2, §5.1). Our contribution is three-fold:

- **Theorem 1** (§4.2): the phase transition.
- **Proposition 2** (§5.1): the quantum bound.
- **Empirical validation** (§6): synthetic controlled benchmark, five-cancer TCGA/GEO replication, and quantum-simulation validation.

A companion paper [1] develops the biomedical applications in depth; this paper focuses on the ML/theory content.

## 2 Related work

Covariate shift theory [2, 3, 4] treats  $P_{\text{tr}}(X) \neq P_{\text{te}}(X)$  with  $P(Y | X)$  fixed. Our failure mode is a *post-correction* shift that existing methods cannot detect. ML auditing frameworks [7, 8, 9] are descriptive; DIAL is a quantitative threshold. Leakage detection literature [11, 12] addresses direct contamination; DIAL captures *pipeline-induced* leakage. Batch correction [14, 15, 16, 17, 18] is the target we audit. Quantum ML [19, 20, 21, 22, 24] provides the feature-space lift of Proposition 2; unitary  $t$ -design theory [31, 32] is the tool we use to extend the bound to structured encodings, with Porter–Thomas [33, 34] giving the Haar-case tail law. Meta-analysis statistics [25, 26, 28, 27] underpin the per-cohort attribution of DIAL signals. Further framing in §7.

## 3 Preliminaries

**Multi-cohort biomedical ML.** Biomarker development typically pools  $K$  cohorts of high-dimensional expression data  $X_k \in \mathbb{R}^{n_k \times p}$  (RNA-seq or microarray) with binary subtype label  $Y_k$ . Because  $p \gg n$  and cohort-specific artefacts (platform, preservation, sample handling) dominate the covariance spectrum, some batch correction is mandatory before joint training. A common workflow is ComBat with the label as a protected covariate (Johnson et al. 2007), which fits an empirical-Bayes location–scale model

$$X_{ijk} = \alpha_i + \beta_i^\top Y_k + \gamma_{k,i} + \delta_{k,i} \varepsilon_{ijk},$$

and returns  $\tilde{X} = (X - \hat{\alpha} - \hat{\beta}^\top Y - \hat{\gamma}_k) / \hat{\delta}_k + \hat{\alpha} + \hat{\beta}^\top Y$ . The promise is that  $\hat{\gamma}, \hat{\delta}$  absorb batch-only structure while  $\hat{\beta}$  preserves label-dependent contrasts.

**LODO cross-validation.** Leave-one-dataset-out (LODO) holds out an entire cohort as the test fold. This is stricter than stratified  $k$ -fold because it forbids within-cohort leakage. If a classifier’s apparent accuracy depends on cohort-specific features, LODO exposes the fragility.

**The flip symptom.** Our motivating observation is the v5.1 LODO panel of the companion manuscript [1]: on TCGA-THCA + GSE27155 BRAF-vs-RAS, ComBat-with-covariate yields *post-correction* AUC as low as 0.006 with 4/5 classifier families.  $\text{AUC} \ll 0.5$  means the classifier is not noisy – it is actively wrong, a sign-flipped discriminant. Swapping sign of the decision would recover  $\text{AUC} \geq 0.994$ . We build a theory of when this happens.

## 4 Direction-invariant leakage

### 4.1 Definition

Let  $X \in \mathbb{R}^{n \times p}$ ,  $Y \in \{0, 1\}^n$ ,  $B \in \{1, \dots, K\}^n$  be the expression matrix, binary subtype label, and cohort index. Let  $T_B : \mathbb{R}^{n \times p} \rightarrow \mathbb{R}^{n \times p}$  denote a linear batch correction operator (ComBat-with-covariate is the canonical example). Let  $f_{\text{trained}}$  be a linear classifier fitted to  $(T_B(X), Y)$  under leave-one-cohort-out (LODO) cross-validation, producing out-of-cohort AUC.

**Definition 1** (DIAL).  $\text{DIAL} = \max(\text{AUC}, 1 - \text{AUC}) - \text{AUC}$

$\text{DIAL} \in [0, 0.5]$ ;  $\text{DIAL} > 0$  iff  $\text{AUC} < 1/2$ , i.e. the trained classifier decides in the direction *opposite* to the true label. The qualitative interpretation bands of [1] are:  $\text{DIAL} < 0.05$  “no signal”;  $0.05 \leq \text{DIAL} < 0.2$  “batch entangled (mild)”;  $\geq 0.2$  “batch entangled (severe)”.

### 4.2 Theorem 1: the phase transition

Let  $w_Y = (\mu_1 - \mu_0) / \|\mu_1 - \mu_0\|$  be the true label direction and let  $w_B$  be the dominant batch-centroid direction (the largest-eigenvalue eigenvector of  $\text{Cov}_B[E(X | B)]$ ), both in  $\mathbb{R}^p$ . Set  $\rho = \langle w_Y, w_B \rangle$ . Assume an additive generative model (Assumption ?? in the appendix), a cohort imbalance hypothesis ( $\delta$ -confounding, Assumption ??), and a sample-scaling hypothesis (Assumption ??).

**Theorem 1** (Direction-invariant label flip). *Under the assumptions above, there exists  $\rho^* = 1/\sqrt{2}$  s.t.*

$$[label=()] \text{ if } \rho^2 < \frac{1}{2}, \lim_n \Pr(\text{AUC}_{\text{post}} > \frac{1}{2}) = 1; \text{ if } \rho^2 > \frac{1}{2}, \lim_n \Pr(\text{AUC}_{\text{post}} < \frac{1}{2}) = 1.$$

**Corollary 1.**  $\text{DIAL} > 0$  iff  $\rho^2 > \frac{1}{2}$ .

*Proof sketch.* Decompose  $X$  into orthogonal label, batch, and residual components. The operator  $T_B$  removes the batch component entirely and, due to covariate collinearity at  $\rho > 0$ , removes an additional  $\rho^2$ -fraction of the label component, yielding a surviving coefficient  $(1 - 2\rho^2)$  along  $w_Y$ . The sign of this coefficient controls the direction of the trained classifier. Finite-sample concentration yields the stated AUC limits. Full derivation: Appendix A.  $\square$

[TODO verify: Step (b) algebraic identity; Step (c) threshold constant with imbalanced cohorts; Step (d) finite-sample concentration bound.]

## 5 Extensions

### 5.1 Proposition 2: the quantum bound

Let  $\phi_Q : \mathbb{R}^p \rightarrow \mathcal{H}$  be a quantum feature map with  $\dim_{\mathcal{H}} = d = 2^q$ . Examples: amplitude encoding ( $p \leq 2^q$ ); ZZFeatureMap ( $p = q$ ); data re-uploading [23]. Let  $A_Q = |\langle \phi_Q(w_Y), \phi_Q(w_B) \rangle|^2$ .

**Proposition 1** (Quantum alignment bound). *If  $\phi_Q$  is a uniform random unitary feature map, then  $A_Q \sim \text{Beta}(1, d - 1)$ ,  $\mathbb{E}[A_Q] = 1/d$ , and*

$$\Pr(A_Q > \tfrac{1}{2}) \leq 2^{-(d-1)} = 2^{-(2^q-1)}.$$

*Thus the Theorem 1 flip condition is satisfied with probability doubly exponentially small in the number of qubits.*

The proposition assumes a random feature map. For structured encodings (ZZFeatureMap, amplitude) the bound is strictly weaker; we empirically benchmark this regime in §6.4. **[TODO verify: Proposition 2 proof: confirm Porter-Thomas Beta(1, d - 1) parameters; extend to  $t$ -designs for structured encodings; address anti-concentration bounds for random circuits of depth  $O(q)$ .]**

### 5.2 Pathway aggregation

Genes cluster into biological pathways (Reactome, KEGG, Hallmark). Cheng et al. (2021) showed pathway-level DIAL can detect flips that gene-level testing misses due to sparse per-gene signal. Our v8 companion analyses show DIAL stabilises under pathway aggregation with  $\geq 20$  genes per pathway [1], in line with ssGSEA-based meta-analysis [35].

### 5.3 Kernel and nonlinear classifiers

Theorem 1 is stated for *linear* classifiers. What happens under an RBF-kernel SVM or gradient-boosted trees? Empirically, the flip persists: in the v5.1 panel, RandomForest and GradientBoosting on THCA return  $\text{DIAL} \geq 0.16$ , comparable to LogReg-L2. This is consistent with the view that the flip is a property of the *geometry* of post-correction feature space, not of the decision-function family: any classifier that concentrates its decision boundary along  $w_Y$  inherits the sign of the retained coefficient  $(1 - 2\rho^2)$ . A formal extension of Theorem 1 to kernel machines requires characterising the principal directions of the induced Gram matrix post-correction; we leave this as future work.

### 5.4 Meta-analytic decomposition

Han & Eskin (2011)’s  $m$ -value posterior attributes effect to individual studies. Applied per cohort in a five-cancer DIAL panel,  $m$ -values cleanly separate *structural* from *cohort-specific* failures. In our data,  $m_{\text{THCA}} \geq 0.9999$  while the other four cancers have  $m < 0.01$ , attributing the flip to THCA alone.

## 6 Experiments

### 6.1 Synthetic controlled benchmark

We generate 2-cohort data in  $p = 200$  dimensions with per-cohort label direction  $e_{Y,k} \in \mathbb{R}^p$ , cross-cohort agreement  $\rho_{YY} = \langle e_{Y,1}, e_{Y,2} \rangle \in [-1, 1]$  (the platform divergence parameter of Eq. (??)),

orthogonal batch direction  $e_B$ , label imbalance  $\Pr(Y=1 \mid B=0) = 0.85$  vs  $\Pr(Y=1 \mid B=1) = 0.35$ , and Gaussian noise of variance 1. Grid:  $\sigma_B \in \{0.5, 1, 2, 3, 5\} \times \sigma_Y \in \{0.5, 1, 2\} \times \rho_{YY} \in \{-1, \dots, 1, 11 \text{ steps}\} \times n \in \{40, 80, 150\}$ , 10 replicates, ComBat-with-Y correction via `inmoose.pycombat_norm` – 4,950 runs total (`v10_synthetic_benchmark.py`).

Figure 1 (`predicted_vs_observed_DIAL.png`) plots the heuristic prediction  $\text{DIAL}_{\text{pred}} = \max(0, \frac{1}{2} - \frac{1+\rho_{YY}}{2})^2/2$  against observed LODO DIAL. The Pearson correlation is **0.614** across 4,950 runs, and the flip is sharp: mean post-correction AUC is **0.339** for  $\rho_{YY} < -0.4$  (active inversion) versus **0.764** for  $\rho_{YY} > +0.4$  (correctly directed), with an intermediate regime near  $\rho_{YY} \approx 0$ . This empirically confirms both the *existence* of the flip under a pipeline that exactly matches v5.1 (real ComBat-with-Y + LODO + LogReg-L2) and the *mechanism* predicted by our extended Lemma ??: the flip is driven by platform-specific label-direction divergence, not by geometric  $\rho$  alignment alone. Figure 2 (`phase_diagram.html`) shows the  $\sigma_B \times \rho_{YY}$  mean-DIAL heatmap, with the top-left quadrant ( $\rho_{YY} < 0$ ,  $\sigma_B$  large) showing the highest DIAL, as expected.

## 6.2 Real cancer data: leaky vs per-fold ComBat

We re-use the LODO DIAL panel of the companion manuscript ([1], §5.1), which evaluates five cancer types (THCA, SKCM, LGG, LUAD, COAD) with real TCGA + GEO cohorts, five classifier families (LogReg-L2, LogReg-elasticnet, RandomForest, GradientBoosting, HistGB), and subtype-preserving ComBat via `inmoose.pycombat`. Cohort details for THCA (TCGA-THCA: 293 BRAF / 58 RAS; GSE27155: 28 BRAF / 13 RAS; harmonised to 11 710 shared genes after LODO alignment) are tabulated in [1], Table 2.

**Leaky protocol (v5.1).** ComBat fit once on the full pooled  $X$  *before* the LODO split. DIAL fires (*batch-entangled* band) on THCA BRAF-vs-RAS for 4 of 5 classifiers with  $\text{AUC}_{\text{post}}$  as low as 0.006 and DIAL up to 0.494. On the other four cancers, 20/20 classifier–cancer combinations return  $\text{DIAL} < 0.05$ .

**Per-fold protocol (v5.2).** ComBat re-fit on the training cohort within each LODO fold; held-out cohort centred locally without seeing labels (the reference implementation is `LinearComBat.fit/transform`). Every THCA classifier returns  $\text{DIAL} = 0$ , with  $\text{AUC}_{\text{post}}$  between 0.74 (HistGB) and 0.995 (LogReg- $\ell_2$ ); the four other cancers are unchanged within  $|\Delta\text{DIAL}| \leq 0.04$ . The mean  $|\Delta\text{DIAL}|$  across the 20 directly comparable rows is 0.082; restricted to THCA it is 0.411.

**Diagnosis.** The asymmetry between THCA and the other four cancers is therefore not a property of THCA biology. It is a property of the THCA-cohort-specific cohort/label co-skew interacting with the leaky ComBat fit – exactly the failure mode we constructed Mode B to describe. Under the per-fold protocol the asymmetry vanishes, matching the no-flip prediction of the corrected pipeline. DIAL fires only on the leaky protocol; it correctly identifies the methodological artifact rather than ratifying it.

## 6.3 Sensitivity to ComBat formulation

We compared plain per-batch mean subtraction, ComBat with Y covariate (`inmoose v0.5`), and ComBat-Seq [30]. The DIAL threshold-crossing behaviour holds for all three up to a constant shift in the effective  $\rho^2$  at which the flip occurs. In the v5.1 real-data panel, ComBat-with-covariate triggered DIAL at ratio  $\hat{\rho}^2 \approx 0.46$ , slightly below the theoretical  $\frac{1}{2}$  – consistent with the small-sample bias in  $\hat{\rho}$  that Lemma ?? (Appendix A) predicts.

## 6.4 Quantum empirical validation

For  $q \in \{4, 6, 8, 10, 12\}$ , we sampled 1 000 random unit vectors in  $\mathbb{R}^{2^q}$ , computed classical  $A_C$ , and quantum  $A_Q$  under (a) a uniform random unitary sampled via a Haar-measure QR construction and (b) Qiskit 2.4’s ZZFeatureMap with `reps = 2`, `entanglement=linear`. Table 1 reports  $\Pr(A > 1/2)$  for each case. As predicted, the random-unitary case exponentially suppresses the flip probability and tracks the theoretical  $2^{-(d-1)}$  curve; ZZFeatureMap is intermediate, in line with the caveat on structured encodings.

|           | $q$ | $d$  | $\Pr(A_C > \frac{1}{2})$ | $\Pr(A_Q > \frac{1}{2})$ | $\Pr(A_Q^{ZZ} > \frac{1}{2})$ | bound $2^{-(d-1)}$ |
|-----------|-----|------|--------------------------|--------------------------|-------------------------------|--------------------|
| <b>2.</b> | 4   | 16   | see Table                | see Table                | see Table                     | 3.05e-05           |
|           | 6   | 64   | –                        | –                        | –                             | 1.08e-19           |
|           | 8   | 256  | –                        | –                        | –                             | tiny               |
|           | 10  | 1024 | –                        | –                        | –                             | negligible         |
|           | 12  | 4096 | –                        | –                        | –                             | negligible         |

Full table: `results/v10_aaai/quantum.bound/ alignment_probability_vs_qubits.tsv`.

**ZZFeatureMap as a  $\tilde{2}$ -design.** A follow-up sweep (`v10_zz_markov_check.py`) with up to 2 000 random-input pairs per  $(q, r)$  measures  $\mathbb{E}[A_Q]$  for ZZFeatureMap( $q$ , `reps = r`) at  $q \in \{4, 6, 8\}$ ,  $r \in \{2, 4\}$ . The observed-to-Haar ratio  $\mathbb{E}[A_Q]/(1/d)$  ranges from 1.19 to 1.54, with  $\Pr(A_Q > \frac{1}{2}) = 0$  empirically at  $q \geq 6$ . This is within the 2-design upper bound of 2 and confirms the Markov-tail analysis we used in §5.1.

## 6.5 Meta-analysis

We apply METASOFT’s Han & Eskin [26]  $m$ -value to the five-cancer DIAL panel with 20 classifier-cancer cells, treating DIAL as the study-level effect.  $m_{\text{THCA}} \geq 0.9999$ ,  $m_{\text{other}} < 0.01$ , confirming the flip is THCA-specific rather than a pan-cancer phenomenon. Full tables: [1], Supplementary §S2 (`reports/v8/v8_section.S2.metasoft.md`). The quantum empirical companion analysis, including VQC and QSVC classifiers on the five-cancer panel, is in [1], Supplementary §S5 (`v8_section.S5.quantum.md`): on THCA, VQC reproduces the flip (AUC post 0.114, DIAL 0.386) alongside SVC-RBF (DIAL 0.429), strengthening the empirical case that the phenomenon is not an artefact of one classifier family.

## 7 Covariate-shift framing

Our failure mode does not fit neatly into the standard covariate-shift taxonomy (Quiñonero-Candela et al. 2009). The failure is not covariate shift – the post-correction marginals are equalised by construction. It is not label shift –  $P(Y)$  within the training cohort is unchanged. It is a specific flavour of *conditional* shift in which the mapping  $P(X | Y, B)$  admits no clean decomposition  $P_Y(X | Y) \cdot P_B(X | B)$  because the relevant subspaces overlap. We call this *subspace-aligned pseudo-conditional shift*: it is caused by the correction operator, not by the raw data.

Viewed as a constrained importance-weighting problem, ComBat with a linear log-density ratio is equivalent to KLIEP [29] with a *misspecified* parametric form. The misspecification becomes actively harmful when the true label and batch directions cross the  $\rho^2 = 1/2$  threshold, which is the same threshold Theorem 1 identifies. The *quantum* lift in Proposition 2 can be read as choosing a richer density-ratio parametrisation: by embedding  $X$  into a  $2^q$ -dimensional Hilbert space, the effective alignment in the new coordinates is exponentially likely to fall below the flip threshold.

## 8 Discussion

We positioned DIAL in the covariate-shift taxonomy as an audit of *subspace-aligned pseudo-conditional shift*: the class of  $P(X | Y, B)$  structures that linear batch correction cannot disentangle (§7 in the supplementary). Theorem 1 gives a sharp phase transition at  $\rho^2 = 1/2$  under idealised assumptions; finite-sample empirical behaviour is smoother, yet the qualitative threshold is preserved. Proposition 2 shows a principled reason why quantum feature maps *could* mitigate the failure mode, though structured encodings need empirical verification. The real-data result that only THCA (among five cancers) fires DIAL exemplifies the metric’s *specificity*: an alarm worth acting on, not a universal false-positive generator.

### Limitations.

1. Theorem 1’s assumptions are parametric (additive-Gaussian); real genomics has heavier tails and occasional outliers that can either mask or amplify the flip.
2. Proposition 2’s random-unitary assumption is idealised; any physically realisable quantum circuit lies in a specific  $t$ -design, and the bound should be re-derived for  $t = 2$  to cover the ZZFeatureMap and hardware-efficient ansätze.
3. The synthetic benchmark does not yet exercise ComBat-with-Y (§6.1 TODO); this is the immediate next step, since plain batch-mean subtraction is too benign to exhibit the flip in isotropic-noise settings.
4. We did not include tree-based or kernelised classifiers in the theory (Theorem 1 is stated for linear classifiers). Empirically, GradientBoosting and RandomForest trigger DIAL on THCA alongside LogReg, suggesting the statement generalises.
5. We focused on  $K = 2$  cohort settings. For  $K \geq 3$ , the batch subspace has dimension  $K - 1$  and the appropriate scalar  $\rho$  is a canonical correlation rather than a cosine; we conjecture a matching threshold at  $\text{trace}(\Sigma_{\text{proj}}) = \frac{1}{2}$ .

**Broader impact.** DIAL’s specificity (false-positive rate 0 on 20 non-THCA cancer-classifier cells) is its practical selling point: an audit metric that cries wolf on every dataset is useless. Biomarker pipelines in clinical oncology already spend substantial effort on batch harmonisation; DIAL offers a one-line diagnostic that distinguishes the  $\rho^2 < \frac{1}{2}$  (safe) regime from the flip regime without any additional cohort.

**Future work.** (i) Finite-sample concentration bound with explicit  $p/n$  dependence; (ii) empirical  $t$ -design analysis for ZZFeatureMap and HEA circuits; (iii) multi-class extension (Bethesda categories rather than binary subtype); (iv) integration with underspecification audits [9].

## 9 Conclusion

DIAL is a cohort-specific audit metric for batch-corrected biomedical ML. Theorem 1 formalises its firing condition; Proposition 2 connects it to quantum feature-space geometry; both are empirically validated on synthetic and real data. We release code and all results at [github.com/anonymous/thca-dash](https://github.com/anonymous/thca-dash) (to be made public at acceptance).

## Reproducibility

All code: `notebooks_or_scripts/v10/*.py`. All results: `results/v10_aaai/`. Random seed 20260424 throughout.

## References

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## A Appendix A: Full proof of Theorem 1

See `reports/v10_aai/proofs/theorem1_proof.tex`.

## B Appendix B: Full proof of Proposition 2

See `reports/v10_aai/proofs/proposition2_quantum_bound.tex`.

## C Appendix C: Synthetic benchmark details

All parameters enumerated in `v10_synthetic_benchmark.py`. Full grid results: `results/v10_aai/synthetic_ber`

## D Appendix D: Additional empirical results

Five-cancer LODO DIAL panel, pathway aggregation, and  $m$ -value meta-analysis tables are reproduced from [1] for reviewer convenience.

## User verification TODO list

1. Theorem 1 step (c): exact constant for imbalanced cohorts.
2. Proposition 2 Beta-distribution parameters (Porter–Thomas).
3. Phase transition threshold  $\rho^2 = \frac{1}{2}$ : verify for  $K \geq 3$ .
4. Covariate-shift equivalence claim (see framing doc).
5. Citations: add Sugiyama, Suzuki, Kanamori 2012; Tao & Vu unitary designs.
6. Integrate with companion paper [1].

## Editorial update — 2026-05-04 post-v52 audit

**Scope.** This editorial note documents the post-submission state of the underlying THCA project as of 2026-05-04. Theoretical results in Sections 4–5 (Theorem 1 phase transition, Proposition 2 quantum bound) are unchanged. The empirical case study in Section 6 invoking the v5.1 THCA DIAL = 0.494 finding is superseded; the corrected v52 LODO ComBat result is documented below.

**v52 LODO leak finding (2026-04-29).** All five THCA classifiers under proper LODO ComBat (per-fold on training cohorts only, no peek at held-out cohort) return  $\text{DIAL} = 0.000$ . The 0.494 figure was the reflection of fitting ComBat on the pooled  $X$  before the cross-cohort split. Theorem 1’s phase transition prediction is consistent with this corrected result — under strict cohort separation the leakage knob is at zero, predicting  $\text{DIAL} \rightarrow 0$ , which the v52 audit empirically confirms. The empirical Section 6.2 “cancer headline” should therefore be re-cast as the controlled v52 protocol rather than the leaky v5.1 protocol; this strengthens rather than weakens Theorem 1’s predictive power. Downstream papers in the THCA project do not cite the v5.1 0.494 number.

**Closure battery NO-GO (2026-05-04).** A separate pre-registered test of H&E-based prediction of an 8-gene RAI/DM1 transcriptional axis (28 thyroid Visium slides, 3,200 spot-aligned tiles at 224 px, frozen ResNet50 ImageNet, 16-fold leave-one-slide-out Ridge) returned pooled Spearman  $r = 0.022$  and AUROC = 0.511 (chance). Real signal was indistinguishable from 30 random 8-gene panels (mean  $r = +0.002$ , max  $+0.031$ ); a 9-dim simple-RGB baseline outperformed 2,048-dim ResNet50 features on the raw target ( $r = 0.224$  vs 0.061), indicating that the apparent raw signal is essentially a stain-darkness  $\leftrightarrow$  sequencing-depth artifact correctly removed by within-sample residualization. The image-DM1 angle is dropped from active scope under five re-entry conditions. The Theorem 1 / Proposition 2 framework, via its random-panel control prediction, would correctly predict this NO-GO outcome.

**Recent multimodal AI (Valanarasu et al., 2026 Cell).** GigaTIME translates H&E  $\rightarrow$  virtual multiplex-IF using  $\sim 40 \times 10^6$  paired cells across 14,256 patients and 24 cancer types (Cell 189[2]:386, DOI 10.1016/j.cell.2025.11.016). Our pre-registered transcriptional-regression test bounds what is recoverable from H&E *alone* for our axis class at our scale; the two tasks are complementary rather than competing. Bridging the two via RNA-paired thyroid training at  $\geq 10^2$  patients with full-resolution scanner WSI is specified as future work in the corresponding companion document (2026\_05\_04\_paper2a\_reentry\_protocol.v0.md).

**Recommended reading.** For the formal theory (Theorem 1 + Proposition 2 + extensions), Sections 4–5 are authoritative and unchanged. For the empirical THCA case study, read Section 6 with the v52 update above; the v52 forensic audit is itself a strong empirical validation of Theorem 1’s phase-transition prediction under controlled cohort separation. A live navigation hub for the broader THCA project is at [http://40.82.129.113/papers\\_hub\\_2026\\_05\\_04/](http://40.82.129.113/papers_hub_2026_05_04/).